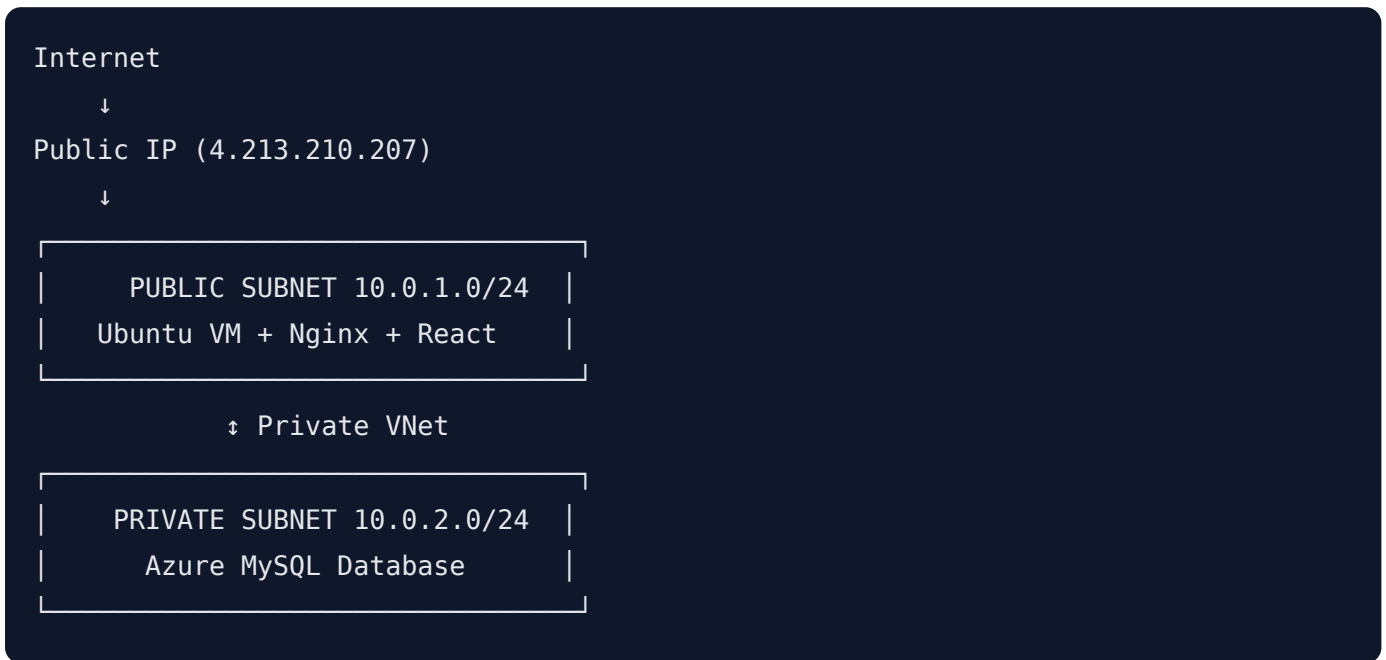


Portfolio Website on Azure — Project Documentation

Architecture



Prerequisites

- Microsoft Azure Account
 - GitHub Account
 - Basic knowledge of Linux commands
-

Step-by-Step Guide

Step 1 — Create Virtual Network

1. Go to **Azure Portal** → **Virtual Networks** → **Create**
2. Fill in:
 - **Name:** portfolio-vnet
 - **Address space:** 10.0.0.0/16

3. Create **2 subnets**:

Subnet Name	CIDR	Purpose
public-subnet	10.0.1.0/24	For VM
private-subnet	10.0.2.0/24	For MySQL

Step 2 — Create Network Security Groups

NSG 1 — For Public Subnet

- **Name:** portfolio-public-nsg
- **Inbound Rules:**

Port	Protocol	Purpose
80	TCP	HTTP Web Traffic
22	TCP	SSH Access

NSG 2 — For Private Subnet

- **Name:** portfolio-private-nsg
- **Inbound Rules:**

Port	Source	Purpose
3306	10.0.1.0/24	MySQL from VM only

Associate each NSG to its subnet.

Step 3 — Create Ubuntu VM

1. Go to **Azure Portal** → **Virtual Machines** → **Create**
2. Fill in:

Field	Value
Name	portfolio-vm
Image	Ubuntu 22.04 LTS
Size	Standard_B1s
Auth	SSH Public Key
VNet	portfolio-vnet
Subnet	public-subnet

3. Download and save the `.pem` key file

Step 4 — Setup the VM

SSH into VM

```
ssh -i your-key.pem shubhamsaini@4.213.210.207
```

Install Required Software

```
# Update system
sudo apt update && sudo apt upgrade -y

# Install Node.js 18
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install -y nodejs

# Install Nginx and MySQL client
sudo apt install nginx mysql-client -y

# Verify installations
node -v
nginx -v
mysql --version
```

Step 5 — Deploy the React App

Clone the Repository

```
cd /home/shubhamsaini
git clone https://github.com/ShubhamCyberStack/portfolio-website.git
cd portfolio-website
```

Fix App.js (Contact Object)

```
nano src/App.js
```

Add contact object mapping:

```
const contact = {
  email: profileData.email,
  linkedin: profileData.linkedin,
  github: profileData.github,
};
```

Install & Build

```
# Install dependencies
npm install

# Build React app
npm run build
```

Step 6 — Configure Nginx

Create Nginx Config

```
sudo nano /etc/nginx/sites-available/portfolio
```

Paste:

```
server {  
    listen 80;  
    server_name _;  
  
    root /home/shubhamsaini/portfolio-website/build;  
    index index.html;  
  
    location / {  
        try_files $uri $uri/ /index.html;  
    }  
}
```

Enable the Site

```
# Remove default site  
sudo rm -f /etc/nginx/sites-enabled/default  
  
# Enable portfolio site  
sudo ln -s /etc/nginx/sites-available/portfolio \  
    /etc/nginx/sites-enabled/  
  
# Set correct permissions  
sudo chmod 755 /home/shubhamsaini  
sudo chmod 755 /home/shubhamsaini/portfolio-website  
sudo chown -R www-data:www-data \  
    /home/shubhamsaini/portfolio-website/build  
sudo chmod -R 755 \  
    /home/shubhamsaini/portfolio-website/build  
  
# Test and restart Nginx  
sudo nginx -t  
sudo systemctl restart nginx  
sudo systemctl enable nginx
```

Verify Website

```
curl -s http://localhost | grep "<title>"
```

Step 7 — Create Azure MySQL Database

1. Go to **Azure Portal** → **Azure Database for MySQL** → **Create**

2. Select **Flexible Server**

3. Fill in:

Field	Value
Name	portfolio-mysql
MySQL Version	8.0
Workload	Development
Admin Username	portfolioadmin
Password	Your strong password

4. Under **Networking**:

- Select ☒ **Private access (VNet Integration)**
- VNet: portfolio-vnet
- Subnet: private-subnet

5. Click **Review + Create**

Step 8 — Load Database Schema & Data

Connect to MySQL from VM

```
mysql -h portfolio-mysql.mysql.database.azure.com \  
-u portfolioadmin \  
-p
```

Load Schema and Seed Data

```
-- Create database
CREATE DATABASE portfolio_db;
USE portfolio_db;

-- Load schema (creates tables)
source /home/shubhamsaini/portfolio-website/db/portfolio_schema.sql;

-- Load seed data (fills tables)
source /home/shubhamsaini/portfolio-website/db/profile_seed.sql;

-- Verify
SHOW TABLES;
SELECT name, title FROM Profile;
EXIT;
```

✓ Final Verification

Check	Command	Expected
Nginx running	<code>sudo systemctl status nginx</code>	active (running)
Website loads	<code>curl http://localhost</code>	HTML response
DB tables exist	<code>SHOW TABLES;</code>	6 tables
Profile data	<code>SELECT name FROM Profile;</code>	Shubham Saini

📁 Project Structure

```
portfolio-website/  
├─ src/  
│   ├─ App.js  
│   ├─ components/  
│   │   ├─ Header.js  
│   │   ├─ About.js  
│   │   ├─ Skills.js  
│   │   ├─ Experience.js  
│   │   ├─ Education.js  
│   │   ├─ Projects.js  
│   │   └─ Footer.js  
│   └─ data/  
│       00─ profileData.js  
├─ db/  
│   ├─ portfolio_schema.sql  
│   └─ profile_seed.sql  
└─ build/          ← generated by npm run build
```

Database Structure

```
portfolio_db  
├─ Profile      → name, title, bio, contact  
├─ Skill        → name, category  
├─ Experience   → position, company, duration  
├─ Education    → degree, institution  
├─ Certification → name, issuer, year  
└─ Project      → name, description, github
```

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